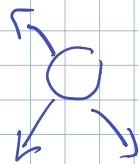


Alberi Binaria



x.key
x.left
x.right
x.parent

Procedure di Visite

Percorrere tutto l'albero e analizzare il contenuto di ogni nodo 1 volta

Pre-order

Pre-visita

In-order

In-visita

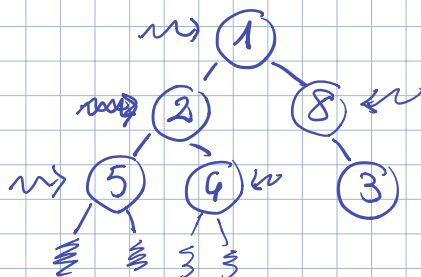
Post-order

Post-visita

Pre-order

- Esaminiamo il nodo print (x.key)
- Procedo analogamente a sinistra
- " " a destra

Esempio Pre-order



1 2 5 4 8 3

```

Pre-order (x) {
    if (x != NIL) {
        print (x.key)
        Pre-order (x.left)
        Pre-order (x.right)
    }
}

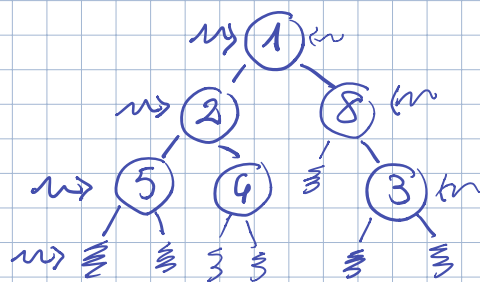
```

Preorder (T.root)

In-order

- Procede a sx del nodo
- Escribo el nodo
- Procede a dx del nodo

Ejemplo In-order



5 2 4 1 8 3

```

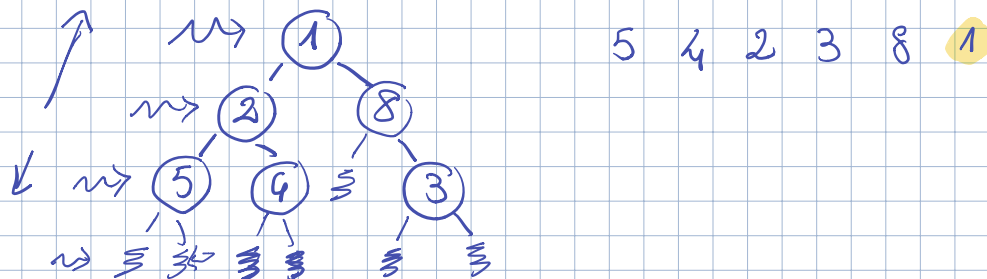
In-order (x) {
    if (x != NIL) {
        In-order (x.left)
        print (x.key)
        In-order (x.right)
    }
}

```

Post-order

- Procedo a sx del nodo
- Procedo a dx del nodo
- Elimino il nodo

Esempio Post-order



Esercizio

Scrivere il codice Post-order

Che complessità hanno?

T con n nodi Pre-Order (T.root)

Pre-order (x) {

if (x != NIL) {

print (x.key)

m $\leadsto$ Pre-order (x.left)

$\leadsto$ Pre-order (x.right)

}

$\Theta(n)$

No in-place

$\Omega(n)$ perché stampa tutti le chiavi;

$O(n)$?

$$T(n) = \begin{cases} \Theta(1) & n=0 \\ T(m) + T(n-m-1) + \Theta(1) & n>0 \end{cases}$$

$$0 \leq m \leq n-1$$

Per Sostituzione dimostro $T(n) = O(n)$

$\Downarrow$

Per induzione su n

$$Hp) \exists c \exists \bar{n} \forall n \geq \bar{n} \quad T(n) \leq c \cdot n$$

Base $n=0$

$$T(0) = 2 \stackrel{??}{\leq} c \cdot 0 \quad \text{No}$$

Base $n=1$

$$T(1) = T(0) + T(0) + b = 2 \cdot 2 + b \stackrel{??}{\leq} c \cdot 1$$

$$\boxed{c \geq 2 \cdot 2 + b}$$

Passo

$$\text{Hplnd}) \quad T(k) \leq c \cdot k \quad \forall k < n$$

$$\S) \quad T(n) \leq c \cdot n$$

$$T(n) = T(\overset{\wedge}{\underset{n}{m}}) + T(\overset{\wedge}{\underset{n}{n-m-1}}) + b$$

$$\leq c \cdot m + c(n-m-1) + b$$

$\stackrel{\S}{\text{Hplnd}}$

$$= \cancel{c \cdot n} - c + b \stackrel{??}{\leq} \cancel{c \cdot n}$$

$$-c + b \leq 0$$

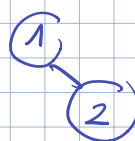
$$\boxed{c \geq b}$$

In-order Post-order $\Theta(n)$

Esercizio

Dato il risultato di una visita (pre-order) è possibile ricostruire T ?

1 2

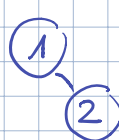


No

Dato il risultato di Pre-order e Post-order
--- T ?

1 2 Pre-order

2 1 Post-order



No

Dato il risultato di Pre-order e In-order?

1 2 5 4 8 3

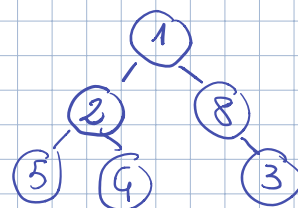
Pre-order

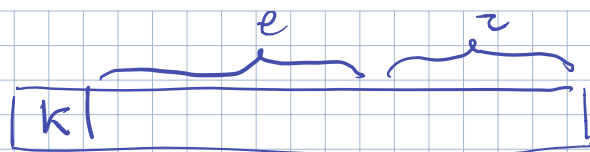
5 2 4 1 8 3

In-order

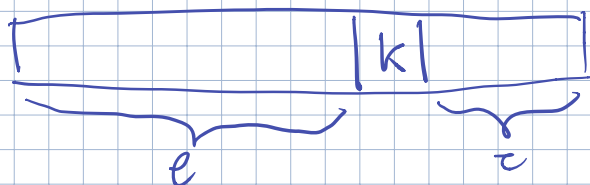
9
x

3
a dx
della root.

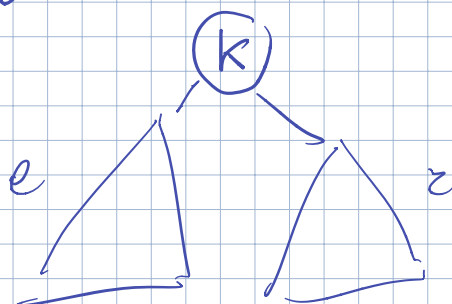




Pre-order



In-order



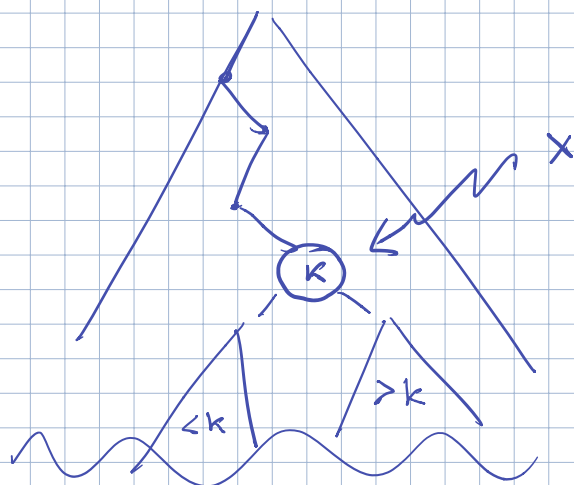
Scrivere
codice
Calcolare
complessità

~ . ~ . ~ .

Alberi Binari di Ricerca (BST)

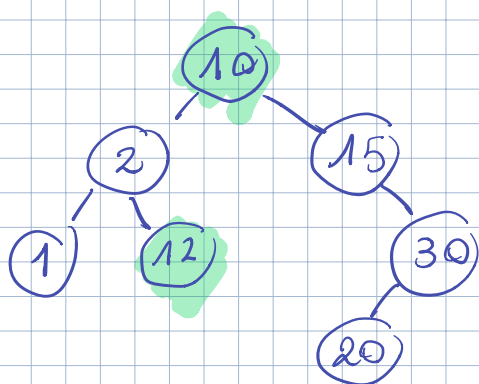
Def T Binary Search Tree (BST)

T albero binario in cui ogni nodo contiene una chiave intera tale che per ogni nodo $x \in T$ tutte le chiavi che si trovano nel sottoalbero radicato a xx di x sono minori delle chiavi $di x$ e tutte $destre$ di x sono maggiori delle chiavi di x

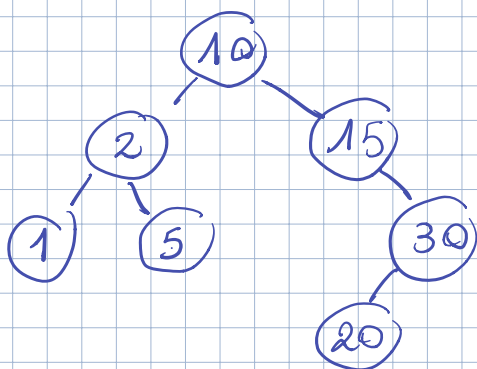


$\forall x \in T \quad \forall y$
 $y \in T_{x.left}$
 $\downarrow$
 $y.key < x.key$
 and
 $y \in T_{x.right}$
 $\downarrow$
 $y.key > x.key$

Esempio



Non è BST

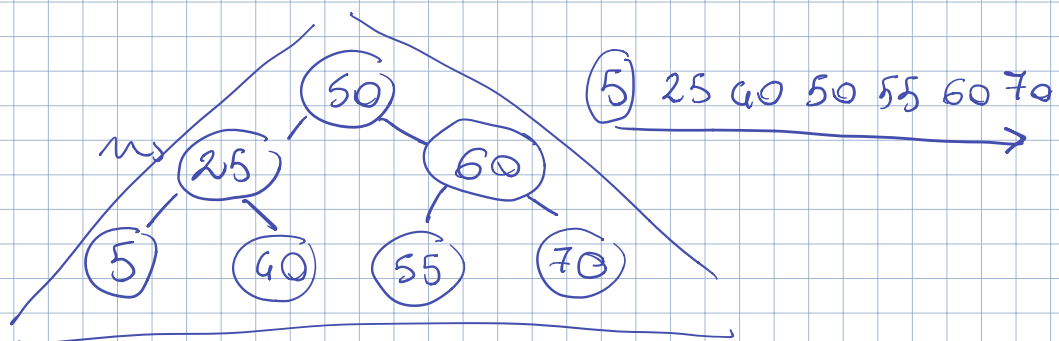
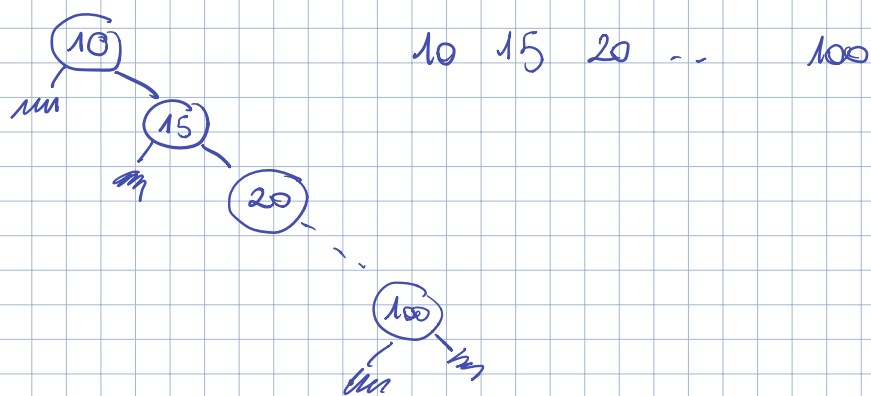


È un BST

Operazioni

- Inserimento
 - Cancellazione
 - Ricerca
- max/min
chiave
predecessore/successore

Esempio



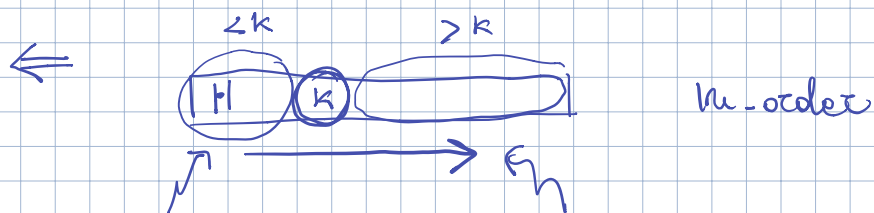
Visite InOrder BST?

Il risultato è un vettore ordinato in ordine crescente

Teo

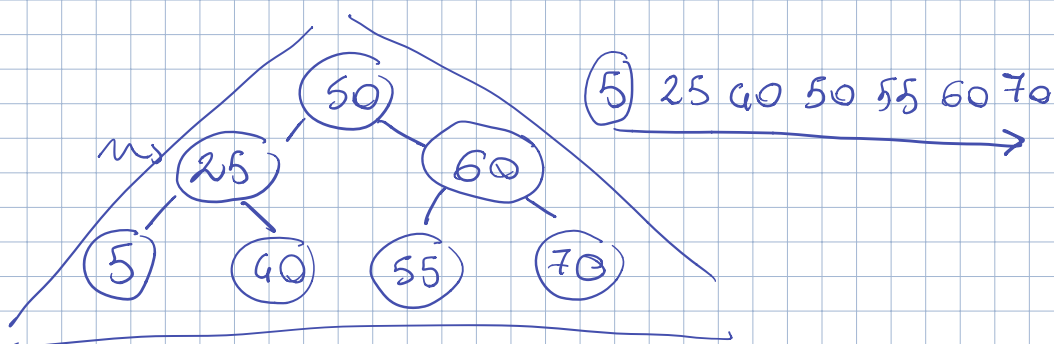
T è un BST $\Leftrightarrow$ in-order (T -cont) è un vettore ordinato

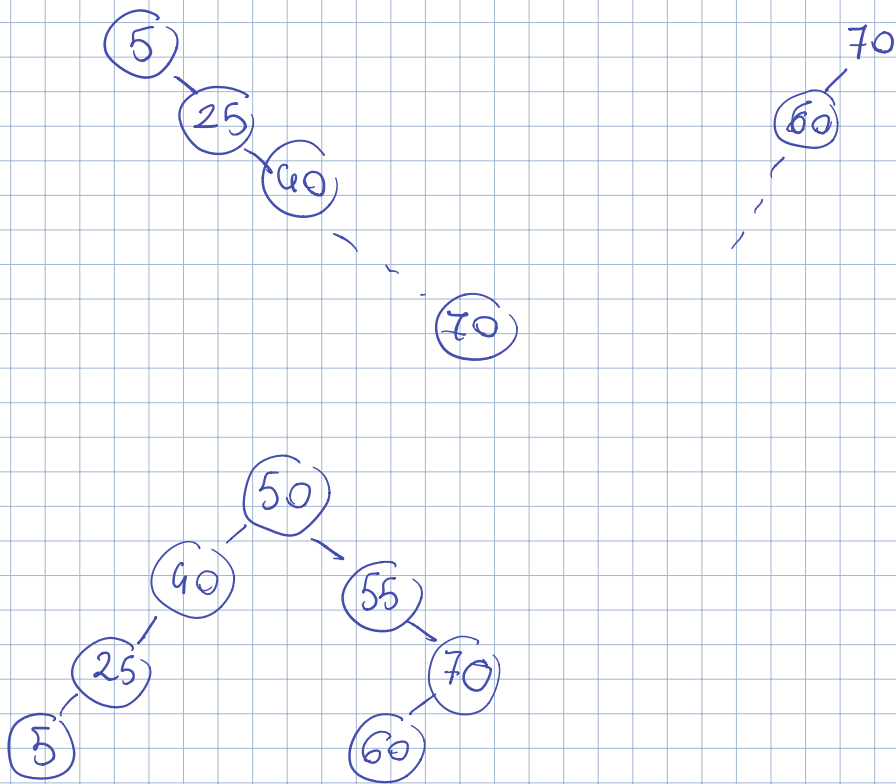
Dim



$\Rightarrow$ Dalla def. di BST

Esempio





Ricerca di Max/Min